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Modeling and Optimization of a Community Microgrid Components

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Abstract

The modeling and optimization of a community microgrid situated in an urban area is presented in this paper. Such communities include places like housing estates, large hospital complexes, large residential complexes, and universities of higher learning. The case study used in this work is a housing estate situated in Cape Town, South Africa. Distributed energy resources used in the study are diesel generators, rooftop solar photovoltaic system, and the electrical load that is segmented into critical and non-critical type. The segmentation of the load allows an intelligent load shedding scheme in order to maintain the stability of the overall system. HOMER modeling software environment is used to model and optimize the community microgrid component sizes for technical and economic feasibility. An optional grid-connected mode as a backup energy system is evaluated against the grid-forming mode of the community microgrid system. Simulation results are obtained and discussed. Conclusion of decision-making about using a grid-forming microgrid system or grid-connected-PV is presented.

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1. Introduction

The increasingly rising cost of grid electricity and the concerns for the environmental impact of primary energy derived from fossil fuels is receiving a concerted effort worldwide [1]. Climate change is now scientifically proven and accepted as being the effect of greenhouse gases trapped in the atmosphere. Carbon dioxide is one of the

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greenhouse gases derivatives of hydrocarbon combustion process. Energy independence can be attained by relying on available local energy resources such as wind and solar resources. Energy systems designed and built based on weather dependent conditions cannot be entirely relied on because of its intermittency and variability characteristic [2]. For better use of solar energy as a dependable resource, a backup energy system is required. This backup energy system can be an electro-chemical energy storage system, such as battery energy system or another energy resource, such as the utility grid or diesel generator that is dispatchable [3].

A microgrid is defined as a group of interconnected loads and distributed energy resources within clearly defined electrical boundaries that acts as a single controllable entity with respect to the grid [4]. In this study, a community microgrid is composed of solar photovoltaic system, diesel generators, energy storage system and composite electrical loads that can be intelligently segregated. A community of housing estate in Cape Town comprising of many family units is designed to have a microgrid system that can reliably and sufficiently cater for the electricity needs at any time of the day for the whole housing complex. Energy storage system is included in the design in order to store excess solar energy produced during the day. This essentially increases the PV system utilization and helps to reduce the magnitude of the imported electricity [5].

The intelligent load shedding scheme is implemented by installing a ripple signal control system for all the geysers in the complex [6]. The ripple signal control system is activated when the total system load tries to overload the diesel generator. The drooping of the generator frequency can be used as an activation signal to switch off all the hot water heating systems of the complex.

The rest of the paper is organized as follows: in the next section, distributed energy resources (DER) is briefly reviewed with emphasis on diesel generation technology, solar photovoltaic, electro-chemical battery energy storage and controllable electric loads as they are the microgrid components used in this study. HOMER capabilities and system model is discussed in section three, results and discussions in given in the penultimate section. Section five concludes the paper.

2. Distributed Energy Resources

Distributed energy resources are composed of energy system, controllable loads and energy storage system. [7] Electrical energy sources such as solar PV, Wind Energy Conversion System (WECS), utility electricity grid, diesel generator, Fuel Cell System (FCS), wave energy harvester, etc. The energy storage system can be electrochemical energy storage system of various technologies, initial energy storage in a flywheel, electrostatic energy in capacitors and supercapacitor, compressed air energy storage system, electromagnetic energy storage in a coil as in Superconducting Magnetic Energy Storage (SMES) system, etc. [8]. Electrical loads can be divided into critical or essential loads and non-critical or deferrable loads. Deferrable electrical loads are the type that can be switched off without causing any noticeable discomfort to the users e.g. hot water heating system or loads that can be shifted to another time of use such as dish washers, washing machines and the likes.

2.1. Diesel generator

A diesel generator consists of a diesel engine and a synchronous generator. The synchronous generator rotor shaft is mechanically coupled to the diesel engine shaft. The diesel engine (also known as a compression-ignition), named after Rudolf Diesel, is an internal combustion engine in which the fuel ignition is caused by the elevated temperature of the air in the cylinder due to mechanical compression. Diesel engines work by compressing only the air. This increases the air temperature inside the cylinder to such a high degree that atomized diesel fuel that is injected into the combustion chamber ignites spontaneously [9]. The magnetically excited rotor converts the input mechanical power to electrical energy through the electromagnetic interaction between the rotor magnetic field and the stator magnetic field. The four main subsystems are the speed governor, the fuel injector, the prime mover and the synchronous generator. The automatic voltage regulator (AVR) is not explicitly shown in the block which is used to regulate the output AC voltage while the governor regulates the mechanical speed for the engine. The mechanical speed of the engine is exclusively related to the frequency of the output voltage waveform.

2.2. Solar Photovoltaic System

A solar PV module is composed of many solar cells connected in a series and parallel combination to fulfill the system requirements of voltage and current rating. Series connection defines the PV module output voltage while parallel connection of the cells determines what the PV module output current is. For understanding the operation characteristics of a solar PV, Fig. 1 below shows the common single diode model of a solar cell electrical equivalent circuit. Equation 1 describes the relationship between the solar cell parameters and variables.

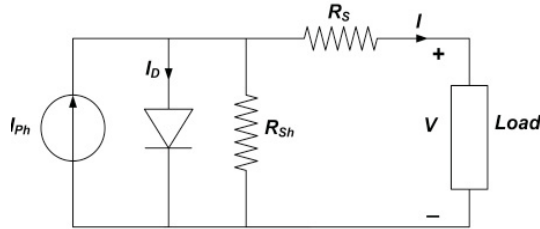


Fig. 1 Single diode model of a solar cell

$$I = I_{ph} - I_D \left\{ \exp \left(\frac{V + IR_s}{A} \right) - 1 \right\} - \frac{V + IR_s}{R_{sh}} \quad (1)$$

Where A is the thermal voltage, R_s and R_{sh} are series and shunt resistances respectively, I_D , I_{ph} and I are the current in the diode when it is directly polarized, the current generated by the solar cell and I the existing current at the terminals of the solar cell respectively.

For efficient utilization of any solar PV module or system, the operating condition at which maximum power is extracted from the cell is tracked and maintained. A power electronics switch-mode DC-DC converter is connected across the PV module output to present an impedance equal to the internal impedance of the PV module. Buck, boost and buck-boost DC-DC converters are some of the switch-mode converters that can be used to implement the maximum power point tracking (MPPT) algorithm decisions [10].

2.3. Battery Energy Storage System

Battery energy storage system converts stored chemical energy to electrical energy. The form of the electricity is DC. A rechargeable battery is able to convert chemical energy to electrical energy and vice versa as well. The most commonly used rechargeable batteries are lead-acid batteries as they are relatively cheap, robust and reliable and have good storage density. The major disadvantage of batteries is their need for the control of charging and discharging processes. This control allows to enlarge the number of charge/discharge cycles and improve the battery life. Therefore, a charge controller is always used to regulate the charge and discharge rate of a battery. An electronic charge controller is responsible for performing this function.

2.4. Electrical Loads

Electrical loads are electronics and electrical appliances that are used in homes, offices, industries, and many other places that are supplied with electrical energy. These appliances mainly convert electrical energy into different useful energy forms such as: light energy (lighting), mechanical energy (motor), heat energy (hot water heating) etc. The use of these appliances can only be predicted and precisely unknown in terms of when and how long it can be switched on. Statistical metrics are usually used to construct the pattern of use. A typical load profile of a household is shown in Fig. 2. It can be seen that there is early morning and evening rise in electricity use. This is expected as generally people wake early in the morning to prepare for work and come back at night to prepare supper and possibly take showers.

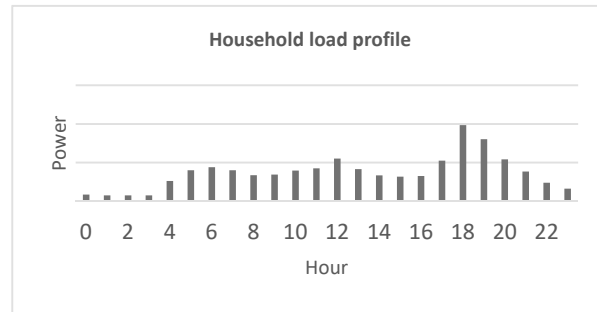


Fig. 2 Residential load profile

3. HOMER Modeling Simulation Optimization and Sensitivity Analysis

HOMER Pro is a commercial software that is used to model, simulate, optimize and sensitize any micro power systems. HOMER Pro is an acronym for Hybrid Optimization of Multiple Electric Renewables. It simplifies the task of assessing designs of both off-grid and grid-connected power systems for a range of applications. HOMER Pro's optimization and sensitivity analysis algorithms make it easier to evaluate the many possible system architectures. The operating principle of HOMER Pro is based on balancing energy produced and consumed in each time step of the year. The simulation process involves determining system configurations that are technically feasible within the defined system constraints. This also estimates the cost of installing and operating the feasible system over the lifetime of the project. The optimization step follows the simulation process. The simulated systems are arranged and filtered according to criteria that are defined. This step is user defined, the default is economic optimization. The sensitivity stage is optional to evaluate the impact of variables that are beyond the modeler control such as fuel costs (such as diesel), wind speed, inflation rate, etc. [11]

Table 1 lists all the costs used for modeling the system under study in HOMER Pro.

Table 1: Modeling component costs

Components	Component costs		
	Capital	Replacement	Operation and Maintenance
PV [12]	US\$ 1000 / 1 kW	US\$ 750	US\$ 55 / year
Batteries [13]	US\$ 1306 / unit	US\$ 1175	US\$ 130 / year
Diesel generator [13]	US 6000 / 10 kW	US\$ 5000	US\$ 0.075 / year
Converter [12]	US\$ 400 / 1 kW	US\$ 300	US\$ 0 / year

4. Results and Discussion

Based on the component costs in Table 1, a simulation of the system in Fig. 3 was carried out using Homer Pro optimizer feature. The system considers a housing estate situated in Cape Town, South Africa, consisting of 30 houses with each requiring 4 kW to meet its critical load demand. The estate noncritical load includes two heat pumps of 6 kW each used to heat and pump water for two swimming pools in the property with each having a size of 7 m x 3.5 m / 25 m³. At the first instance, the system is simulated without the grid (see Fig. 3), thereafter an optional grid-connected mode as a backup energy system is evaluated against the grid-forming mode of the community microgrid system.

Assumptions made during the simulation include an annual discount rate of 8 %, an inflation rate of 2 %, project lifespan of 25 years and capacity shortage fraction of 0 %.

The optimization results of the community microgrid component sizes for technical and economic feasibility are displayed in Table 2 and Table 3. The community microgrid size is such that there is a combination of 129 kW of PV, 126 strings of batteries (each battery has a nominal voltage of 4 volts, a nominal energy capacity of 7.55 kWh, a capacity of 1350Ah, a round trip efficiency of 80-85% and a lifetime around 12 to 20 years. The minimum state of

charge is assumed to be 80%), 140 kW of a diesel generator and 60.9 kW of a converter. The levelized Cost of Energy of this system is 0.509 US\$/kWh, while the Net Present Cost is US\$ 1.64 M (see Table 3).

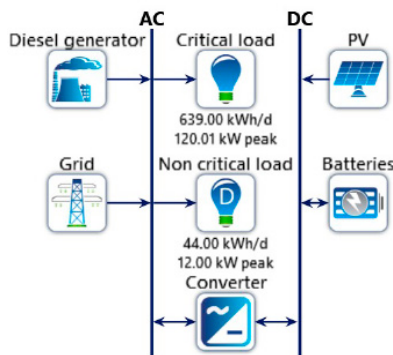


Fig. 3 Housing estate Homer model

Table 2 Community microgrid component sizes

Component	Size	Unit
PV	129	kW
Batteries	126	strings
Diesel generator	140	kW
Converter	60.9	kW
Cost of Energy (COE)	0.509	US\$
Dispatch strategy	HOMER cycle charging	

Table 3 Community microgrid component economies

Net Present Cost					
	Capital (US\$)	Operating cost (US\$)	Replacement cost (US\$)	Salvage (US\$)	Total (US\$)
PV	388,461	16,739	0.00	0.00	405,200
Batteries	164,556	211,753	117,894	-28,235	465,968
Diesel generator	84,000	182,162	56,458	-12,800	743,944
Converter	18,280	0.00	7,756	-1,460	24,576
System	655,297	410,654	182,108	-42,495	1.64M

An optional grid-connected mode as a backup energy system (see Fig. 3) is evaluated against the grid-forming mode of the community microgrid system in terms of the comparison of the overall cost of both systems. The grid data used in the simulation are provided in Homer Pro software and shown in Table 4.

Table 4 Grid data

Grid extension input data	
Capital costs (US\$/km)	8,000
Operation and maintenance costs (US\$ per year per km)	160
Grid energy price (US\$/kWh)	0.1

The break-even distance expresses the distance from the grid that allows the decision between a grid extension and an off-grid system to be made based on both systems Net Present Cost. Fig. 4 shows the Total Net Present Cost of the community microgrid system as a function of the grid-forming cost. Considering the economical point of view, the community microgrid is more cost-effective if that distance is greater or equal to 130.84 km (see Fig. 4) and for distances lesser than that value, the house estate grid connection will be more realistic.

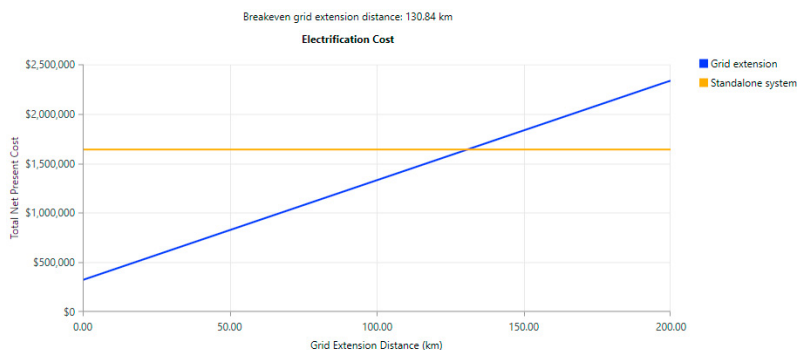


Fig. 4 Community microgrid and grid extension cost comparison

5. Conclusion

This study investigated the modeling and optimization of a housing estate microgrid component sizes for technical and economic feasibility. The levelized cost of energy of the community microgrid in an off-grid form is estimated to US\$ 0.509. An optional grid-connected mode as a backup energy system was compared against the grid-forming mode of the community microgrid system. The comparison between both systems revealed that based on economic considerations, the community microgrid in an off-grid form will be less realistic if the breakeven distance is lesser than 130.84 km.

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